SUDIP CHAKRABORTY, Ph.D.

SOFTWARE ENGINEER | MACHINE LEARNING SYSTEMS | WEATHER FORECASTING & DATA ENGINEERING

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ENGINEERING PROFILE

Scientific software engineer and machine learning researcher with 10+ years building computational tools and data-intensive workflows for satellite observations, atmospheric science, and geospatial prediction. Led software development for NASA JPL's cloud-based Air Quality Analytics Collaborative Framework (AQACF) using Python, AWS EC2/EKS, Docker and JupyterHub. Hands-on experience with forecasting models, data assimilation (Kalman filtering, 3D-Var/4D-Var), PyTorch/CNN/Transformer workflows, GPU-based HPC training, Python package delivery, containerized inference APIs, and large multidimensional data pipelines. Builds reproducible software connecting observational inputs, model inference, diagnostics and downstream scientific users.

CORE TECHNICAL CAPABILITIES

Software & Infrastructure: Python, Git/GitHub, SQL, shell, Docker, FastAPI, REST/JSON model serving, AWS EC2, AWS EKS/Kubernetes-hosted JupyterHub, cloud scientific compute, reproducible environments, reusable pip packages, error debugging and pipeline QA.

Data Engineering & Scientific Computing: NumPy, pandas, xarray, netCDF4, GDAL, rasterio, HDF/NetCDF/GeoTIFF, data ingestion, data quality control, spatial/temporal interpolation, ETL, raster/vector fusion, time-series processing, scalable gridded-data analysis.

ML, Forecasting & HPC: PyTorch, CNNs, Transformers, Mamba/state-space models, random forests, XGBoost, regression, uncertainty estimation, GPU experiments on TACC Lonestar6 A100, model evaluation, forecasting and inference; vLLM model serving on Apple Silicon.

Atmospheric Data & DA: Kalman filtering; variational DA (3D-Var/4D-Var); background/observation error covariance (B/R); observation operators, innovations, radiance vs. retrieval assimilation concepts; ERA5, ECMWF/NCEP weather data, winds, humidity, temperature, satellite L0/L1/L2 products.

SELECTED SOFTWARE ENGINEERING & ML SYSTEMS OWNERSHIP

NASA JPL — Air Quality Analytics Collaborative Framework (AQACF) | Lead Software Developer

- Led engineering of a cloud-based atmospheric data analytics and prediction environment, connecting heterogeneous satellite/environmental inputs, computational analysis, visualizations, and ML research workflows for collaborators.
- Built Python data-processing workflows for multidimensional atmospheric and satellite data, including ingestion, alignment, QC, spatial/temporal resampling, statistical diagnostics, reusable analyses and interpreted outputs.
- Supported AWS EC2 compute and AWS EKS-hosted JupyterHub/Open Climate Workbench infrastructure; packaged scientific/ML environments using Docker to improve reproducibility and portability.
- Integrated CAL2035 exposure and health-impact analytics into shared scientific workflows; collaborated across research and software users to deliver usable data products and troubleshoot processing challenges.

ML Forecasting & HPC — QGLLM, IcePatchNet and Scientific Model Workflows

- Designed QGLLM Arctic sea-ice concentration forecasting experiments combining satellite and ERA5 features (SST, winds, albedo, AOD, radiation and atmospheric fields) with spatial/temporal attention, Transformer and Mamba approaches.
- Built data preparation, tokenization, temporal train/test splits, evaluation and ablation workflows for high-dimensional gridded forecasting; used TACC Lonestar6 A100 GPU infrastructure for training experiments.
- Developed IcePatchNet, a pip-installable patch-CNN forecasting package supporting 1-, 7-, and 14-day sea-ice concentration forecasts; engineered repeatable preprocessing, training/inference and verification workflows.
- Developed CMAD, a pip-installable modular framework for rare-event anomaly detection and prediction in spatial sea-ice observations; complementary CMAD-Vis work enables interactive analysis of anomalies.

Deployment, Model Serving & Reusable Software

- Built and ran a Dockerized FastAPI prediction service for sea-ice concentration using a serialized trained model, structured request schema and local HTTP endpoint; maintained app, dependencies, Dockerfile and versioned source code.
- Installed and operated vLLM with a Qwen model for local inference on Apple Silicon; exercised inference requests, streaming responses and decoding controls as part of practical ML serving work.
- Created reusable, documented Python tools and scientific processing components used for research collaboration, algorithm evaluation, debugging and reproducible delivery.

FORECASTING, DATA ASSIMILATION & OBSERVATIONAL SYSTEMS

Numerical Weather, Remote Sensing & Data Assimilation

- Applied estimation and physical-modeling knowledge encompassing Kalman filters, 3D-Var and 4D-Var, observation operators, innovations, observational/background uncertainty, and multivariate geophysical constraints.

- Processed Level 0 raw instrument data concepts and Level 1 calibrated/geolocated and Level 2 retrieved satellite products, including MODIS, VIIRS, MLS, CloudSat, CALIPSO, SMAP, ECOSTRESS, ICESat-2, Landsat and Sentinel datasets.
- Engineered geospatial and meteorological workflows to integrate reanalysis, satellite data and spatial observations; addressed inconsistent resolution, missing values, quality flags, geolocation, time matching and cross-source comparisons.

Applied Atmospheric Prediction & Environmental Data Engineering

- Developed/used machine learning and statistical models for drought and hydroclimate prediction, atmospheric aerosol transport and environmental forecasting; collaborated with Texas Water Development Board and public-sector partners.
- Analyzed wildfire smoke and aerosol transport with satellite/reanalysis datasets, atmospheric features and statistical/ML methods; built spatial diagnostics and quantitative validation analyses.
- Used graph-based analytics (Delaunay triangulation, spatial link structures and Monte Carlo significance) to characterize interconnected sea-ice/ice-shelf variability and explain model-derived relationships.

PROFESSIONAL HISTORY

Research Assistant Professor, Information Systems | UMBC | 2023–Present

Research Scientist II / Postdoctoral Researcher | NASA Jet Propulsion Laboratory | 2018–2023

Postdoctoral Researcher | UCLA | 2017–2018 • Postdoctoral Researcher | UT Austin | 2016

EDUCATION & SELECTED OUTPUTS

Ph.D., Atmospheric Physics | The University of Texas at Austin

Gundu, S., Chakraborty, S., Janeja, V.P. “QGLLM: Quant-Generative Large Language Model for Sea Ice Prediction.” Submitted to HiPC 2026.

Devnath, M.K., Chakraborty, S., Janeja, V.P. “Modeling Heterogeneity across Varying Spatial Extents.” ACM SIGSPATIAL 2025.

Software: CMAD and IcePatchNet (pip-installable Python packages); NASA/JPL AQACF; containerized FastAPI model API.